Monthly Intelligence Report

Edition 01 — The Mochovce Nuclear Corridor

Where four VVER-440 units, an automotive backbone and an east-west economic gradient meet. Inaugural edition · July 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

1,516

SEPS + ZSD/SSD/VSD heritage

GRID LINES MAPPED

~1,636

400 / 220 / 110 kV + MV

MC SIMULATIONS

15.16 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). For Slovakia, its engine processes 32,000+ source data points per run, executes 15.16 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 360+ million arithmetic operations — producing 90,000+ output data points with full uncertainty quantification across 1,516 substations and ~1,636 grid lines.

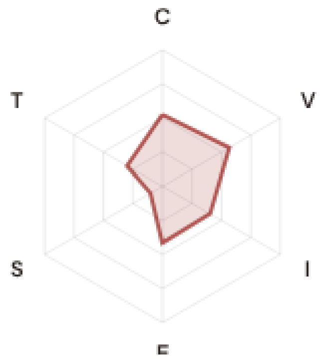
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources — SEPS network statement, ÚRSO quality reports, ZSD / SSD / VSD filings, ŠÚ SR RegDat, SHMÚ meteorology + hydrology, ÚJD SR nuclear oversight, SK-CERT cyber posture — making the methodology fully auditable and reproducible. Slovakia is the second member of the CEE-South-2026 cohort, deployed in dual-drop with Slovenia on the 9 July 2026 anchor.

Substation in Focus

TS-953

SK010, SK010 · 110 kV

0.401	Medium	0.050	49th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

C Continuity: 0.524 / 0.30 (175%)

I Infrastructure: 0.403 / 0.25 (161%)

S Saturation: 0.100 / 0.20 (50%)

V Voltage: 0.567 / 0.10 (567%)

E Economic: 0.415 / 0.10 (415%)

T Transition: 0.299 / 0.05 (598%)

MODIFIERS

R3 Consequence: 0.952 ↓ dampens

R4 Graph Criticality: 0.938 ↓ dampens

R6a Restoration: 1.057 ↑ amplifies

R6b Seismic: 1.123 ↑ amplifies

R7 Digital Readiness: 0.966 ↓ dampens

This month's spotlight — automatically selected for June 2026 — is **TS-953** in SK010, SK010. With an R_final of 0.401 (Medium band, 49th fleet percentile), its risk profile is dominated by the T (Transition) component at 598% of its weight ceiling, followed by V (Voltage) at 567%. Modifiers collectively shift R_base by 6.1%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Slovakia's Act 366/2024 NIS2 transposition + NBÚ competent-authority designation require.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

27

High/Critical + R7 < median

AVG R7 MODIFIER

0.9799

Range [0.99, 1.03]

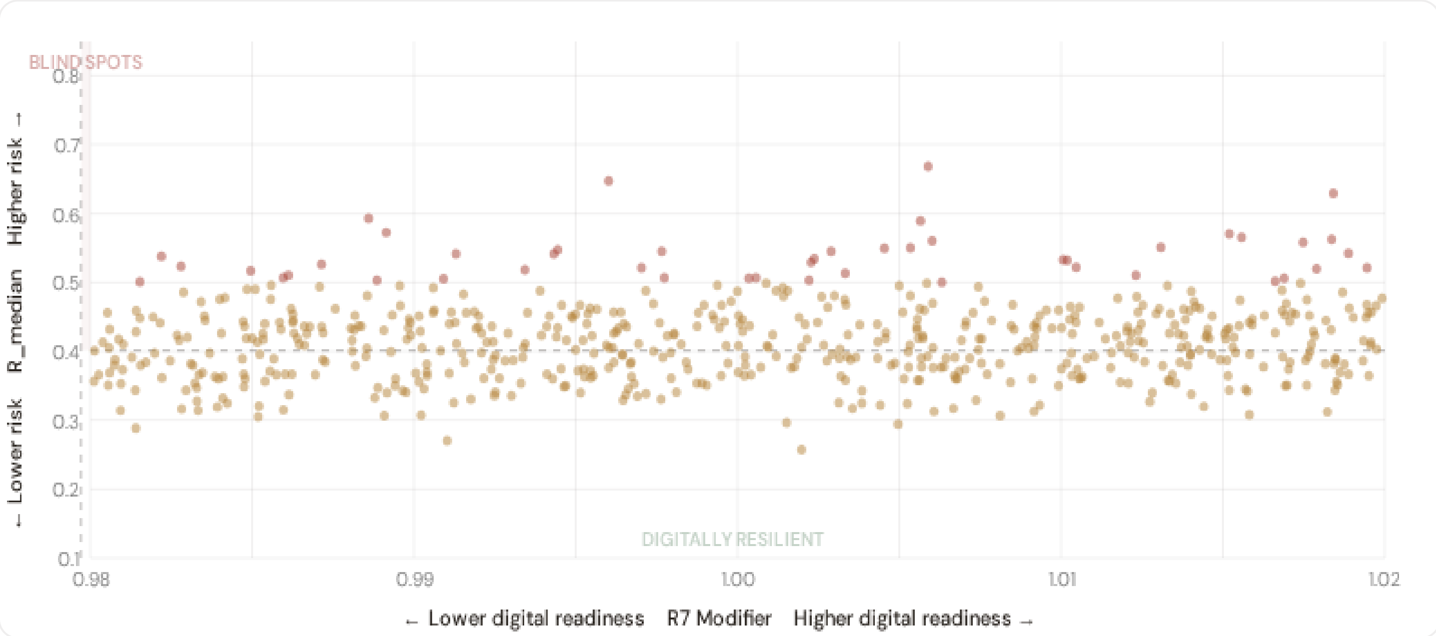
HIGH-CONSEQUENCE SUBS

379

25.0% · R3 ≥ 1.04

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **27 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9797). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **SK041 (8), SK042 (6), SK031 (4)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh

Bratislava metropolitan core + capital-intensive industry (steel, chemicals, refining); SAIDI-sensitive automotive Tier-1/2 cluster Trnava–Nitra

379 substations (25.0%) High/Critical: **37** (9.8%) Avg R: **0.426** Avg E: **0.452** / 0.10

Low 0 Med 342 High 37 Crit 0

🔧 Industrial-Secondary / Heavy-Industry

Est. VoLL: €8–15/kWh

Košice + Žilina industrial corridors; U.S. Steel Košice + Volkswagen Slovakia + Kia Slovakia + Volvo Košice

379 substations (25.0%) High/Critical: **34** (9.0%) Avg R: **0.416** Avg E: **0.452** / 0.10

Low 0 Med 345 High 34 Crit 0

🚗 Industrial-Major / Automotive Corridor

Est. VoLL: €4–8/kWh

Mid-tier industrial substations across Trnava + Trenčín + Bratislava krajs; SME-dense automotive supply-chain feeders

379 substations (25.0%) High/Critical: **12** (3.2%) Avg R: **0.395** Avg E: **0.450** / 0.10

Low 0 Med 367 High 12 Crit 0

🌾 Light-Rural / East-Lagging

Est. VoLL: €1–3/kWh

Prešov + Banská Bystrica + rural eastern krajs; agricultural + light-residential SAIDI bands

379 substations (25.0%) High/Critical: **5** (1.3%) Avg R: **0.381** Avg E: **0.449** / 0.10

Low 1 Med 373 High 5 Crit 0

Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **37 substations** combine capital-intensive economic fabric (R3 ≥ 1.04) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **SK042 (87), SK010 (73), SK031 (72)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 6.9%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 NUTS-3 Kraj Digital Readiness Ranking

Kraje ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions + SK-CERT maturity). Cross-referenced with average R_median to identify kraje combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 NUTS-3 REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	SK032 ⚠ high R	0.9755
2	SK010	0.9768
3	SK021	0.9779
4	SK031	0.9783
5	SK022 ⚠ high R	0.9803
6	SK041 ⚠ high R	0.9813
7	SK042 ⚠ high R	0.9821
8	SK023 ⚠ high R	0.9868

NUTS-3-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **SK032** has the lowest average R7 (0.9755), while **SK023** leads at 0.9868 — a spread of 0.0113 across the R7 range. **1 nuts-3 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9799, median = 0.9797; 0 substations classified HIGH cyber-exposure; 27 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **18 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

18

95 variables

WEEKLY / MONTHLY

3

High-frequency feeds

QUARTERLY / ANNUAL

10

Regular refresh cycle

STATIC / DERIVED

0

Standards + scenarios

Data Source Registry — June 2026

d01_seps	SEPS — Slovenská elektrizačná prenosová sústava (TSO)	DAILY	C2,C3,I1,I2,I3,R4	14 vars
d03_shmu	SHMÚ — Slovenský hydrometeorologický ústav	DAILY	I1,I2,I3,R6c_flood	8 vars
d14_okte	OKTE — SK day-ahead + intraday operator	DAILY	T1,E1	4 vars
d05_osm	OSM Overpass — grid topology	MONTHLY	I1,I2,I3,R4	8 vars
d06_sazp	SAŽP — Slovenská agentúra životného prostredia	MONTHLY	R5_corrosion,I8	5 vars
d07_copernicus	Copernicus ERA5 + CMIP6	MONTHLY	R6_climate	4 vars
d09_sk_cert	SK-CERT (operated by NBÚ) + GovCERT.SK	CONTINUOUS	R7_cyber	5 vars
d02b_dso	3 regional DSOs (ZSD / SSD / VSD)	QUARTERLY	C1,V1,V2,I2,I3	18 vars
d04_susr	ŠÚ SR — Štatistický úrad Slovenskej republiky	QUARTERLY	E1,E2,S1,S2,S3	16 vars
d04b_nbs	NBS — Národná banka Slovenska (central bank)	QUARTERLY	E1,E2	4 vars
d08_ujd_sr	ÚJD SR — Úrad jadrového dozoru (nuclear safety)	QUARTERLY	R6_seismic,I3	4 vars
d02_urso	ÚRSO — Úrad pre reguláciu sieťových odvetví	ANNUAL	C1,C2,V1,V2,T1	12 vars
d10_nbu	NBÚ — Národný bezpečnostný úrad (NIS2 competent authority)	ANNUAL	R7_cyber	3 vars
d11_entsoe	ENTSO-E TYNDP + Transparency Platform	ANNUAL	I1,I2,T1	6 vars
d12_eurostat	Eurostat — EU-27 NUTS-3 benchmarks	ANNUAL	E1,S1,T1	8 vars
d13_se	Slovenské elektrárne (Mochovce + Bohunice V2)	ANNUAL	I1,I3,R6_seismic	5 vars
d15_iea_oecd	IEA + OECD — energy benchmarks	ANNUAL	T1,E1	6 vars
d03b_sgudz	ŠGÚDŠ — Štátny geologický ústav D. Štúra	MULTI-YEAR	R6_seismic	4 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Monthly (3 sources) — Grid load + procurement data

OSM Overpass · SAŽP (air quality) · Copernicus ERA5/CMIP6

Quarterly (4 sources) — DER registry + business + macro

DSO filings (ZSD/SSD/VSD) · ŠÚ SR · NBS · ÚJD SR (nuclear)

Annual (6 sources) — Core grid quality + demographics

ÚRSO · ENTSO-E TYNDP · Eurostat · NBÚ · Slovenské elektrárne · IEA/OECD

C.2 Fleet Score Snapshot

FLEET SIZE	MEDIAN R	HIGH + CRITICAL
1,516 substations scored	**0.401** P5=0.315 · P95=0.505	**88** 5.8% of fleet

HIGH CONFIDENCE

100%

1516 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 1 (0.1%) · Medium 1427 (94.1%) · High 88 (5.8%) · Critical 0 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **1,516 substation scores remain unchanged** this edition. The fleet median R of 0.401 (mean 0.405) reflects a distribution skewed toward the lower bands, with 0.1% in Low and 94.1% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

9 changes tracked across P0, P1, and P2 releases

SK-S28-9	ENHANCED	Greenfield thin-shell rollout — 8 pages authored directly on Phase-2 architecture (no fat-HTML predecessor)	KB §65
SK-S28-8	NEW	Section G deepDives rotation — 8 SK kraje (Mochovce Unit-4 commissioning anchor on Nitriansky kraj)	intelligence G
SK-S28-7	ENHANCED	Edition patcher anchored at FIRST_REFRESH 2026-07-09 (CEE-South-2026 dual-drop with Slovenia)	intelligence

SK-S28-6	NEW	ESG report — slovakia entry in COUNTRY_SOURCES (14 SK-specific references)	esg-report
SK-S28-5	NEW	R3 4-tier calibration — Capital / W-Auto / TN-KE / BB-PO (vs SI 3-tier; reflects east-west GDP gradient)	methodology
SK-S28-4	NEW	R6 modifier set — Pieniny seismic (PO/KE) + Danube/Tisza flood (BA/NR/KE/PO) + Tatra windstorm 2004 anchor	methodology
SK-S28-3	DATA	d05_osm LIVE — 1,516 substations + 1,636 power lines via ISO3166-1=SK area filter	methodology
SK-S28-2	DATA	3 regional DSOs (ZSD/SSD/VSD) + SEPS TSO + ÚRSO regulator + ÚJD SR nuclear safety wired	methodology
SK-S28-1	ENHANCED	KB v25 §65 — Slovakia inaugural onboarding (CEE-South-2026 cohort member 2 of 3)	KB §65

SECTION D — MONTHLY DEEP-DIVE

Nitriansky kraj — The Mochovce Nuclear Corridor

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation covers all 8 Slovak NUTS-3 kraje plus repeat themes: **Ed. 01** Nitriansky — Mochovce nuclear corridor · **02** Bratislavský — capital load + Danube · **03** Trnavský — Bohunice V2 + PSA · **04** Trenčiansky — Považská corridor + Nováky coal exit · **05** Žilinský — Tatras + Kia · **06** Banskobystrický — central highlands · **07** Prešovský — Pieniny seismic + east lagging · **08** Košický — U.S. Steel + SK-UA Mukacheve corridor · **09–12** repeat cycle.

D.1 Why Nitriansky kraj, Why Now

CORRIDOR SUBSTATIONS
116
2 EHV · 114 HV

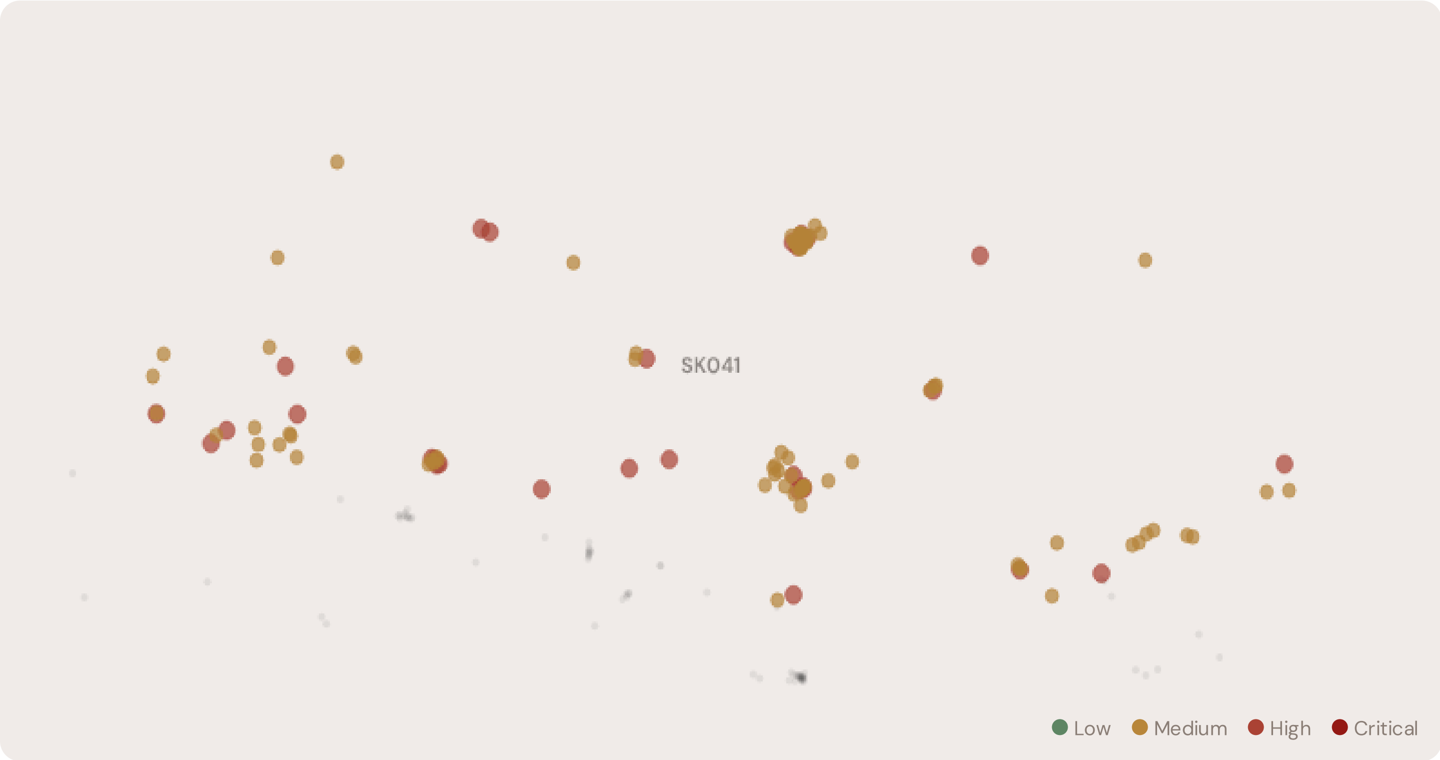
HIGH BAND
30
25.9% of corridor

CORRIDOR MEDIAN R
0.458
Fleet median: 0.401

WORST NUTS-3 KRAJ
SK041
Avg R: 0.469 · 116 substations

The Nitriansky kraj — Mochovce nuclear corridor hosts **116 substations** across 1 nuts-3 regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **30 substations (25.9%)** are classified High, with only **0** in the Low band. 89% of corridor substations score above the national fleet median of 0.401. The corridor median R of 0.458 reflects the local risk profile. Nitriansky kraj (SK023) is the country's nuclear concentration zone. All four Mochovce VVER-440/V-213 units sit at a single site near Levice — Units 1+2 commercial since 1998/2000, Unit 3 since October 2023 after roughly a decade of construction delays, Unit 4 in commissioning with hot functional testing from February 2025 and commercial operation expected in the first half of 2026. Post-Unit 4 the site contributes ~1,880 MWe, lifting Slovakia's nuclear share above 60% and making the country one of the most nuclear-dependent grids in the EU. The Industrial-Major R3 tier (1.04) reflects the kraj's automotive backbone (Jaguar Land Rover Nitra) layered on top of the nuclear cluster — a combination that drives high consequence multiplier sensitivity to any single-site outage. ÚJD SR safety oversight, ENTSO-E reserve obligations, and SEPS balancing cascade across all 3 DSOs simultaneously in a Mochovce trip scenario.

D.2 Corridor Map and Scoring



SUBSTATION	KRAJ	R_FINAL	BAND	C	V	I	E	S	T	ALERT
SK_101447	SK041	0.668	High	0.712	0.711	0.542	0.541	0.100	0.433	C, V
Lemešany	SK041	0.647	High	0.648	0.696	0.643	0.524	0.100	0.593	V
SK_101465	SK041	0.629	High	0.631	0.461	0.665	0.626	0.100	0.602	I
TS0023-0103	SK041	0.612	High	0.628	0.583	0.579	0.616	0.100	0.637	
Elektrická stanica Prešov 3	SK041	0.593	High	0.653	0.578	0.614	0.610	0.100	0.405	C
SSM Strážske	SK041	0.591	High	0.696	0.471	0.689	0.610	0.100	0.602	C, I
SK_100316	SK041	0.590	High	0.563	0.666	0.683	0.431	0.100	0.632	V, I
SK_101244	SK041	0.572	High	0.557	0.451	0.630	0.477	0.100	0.475	
SK_101352	SK041	0.570	High	0.706	0.465	0.525	0.600	0.100	0.417	C
TS 0591-0064	SK041	0.565	High	0.635	0.337	0.633	0.558	0.100	0.665	T

SUBSTATION	KRAJ	R_FINAL	BAND	C	V	I	E	S	T	ALERT
… 106 additional substations — *explore full corridor in Digital Twin →*										

D.3 Component Analysis

COMPONENT AVERAGES — NITRIANSKY KRAJ — MOCHOVCE NUCLEAR CORRIDOR VS FLEET

C — Continuity 0.538 vs 0.475 (+13%)

I — Infrastructure 0.531 vs 0.466 (+14%)

S — Saturation 0.100 vs 0.100 (0%)

V — Voltage 0.519 vs 0.460 (+13%)

E — Economic 0.512 vs 0.451 (+14%)

T — Transition 0.522 vs 0.459 (+14%)

MODIFIER AVERAGES — NITRIANSKY KRAJ — MOCHOVCE NUCLEAR CORRIDOR VS FLEET

R3 Consequence	1.013	fleet 0.999	↑
R4 Graph Crit.	0.980	fleet 0.980	↓
R6a Restoration	1.027	fleet 1.022	↑
R6b Seismic	1.057	fleet 1.049	↑
R7 Digital Read.	0.981	fleet 0.980	↓

The dominant risk driver across the Nitriansky kraj — Mochovce nuclear corridor corridor is **T (Transition)**, averaging 0.522 against a fleet average of 0.459 — occupying 1044% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.519 vs fleet 0.460 (519% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.013 (fleet: 0.999), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.057, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 NUTS-3 Kraj Breakdown

NUTS-3 RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

SK041 116 subs · avg R = 0.469 · H:30 M:86

The nuts-3 regions-level breakdown reveals significant intra-regional variation. **SK041** leads with a mean R of 0.469 across 116 substations, while **SK041** scores 0.469 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Nitriansky kraj — Mochovce nuclear corridor is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

1 Nitriansky kraj — Mochovce nuclear corridor substations score R ≥ 0.650, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the SK041 sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

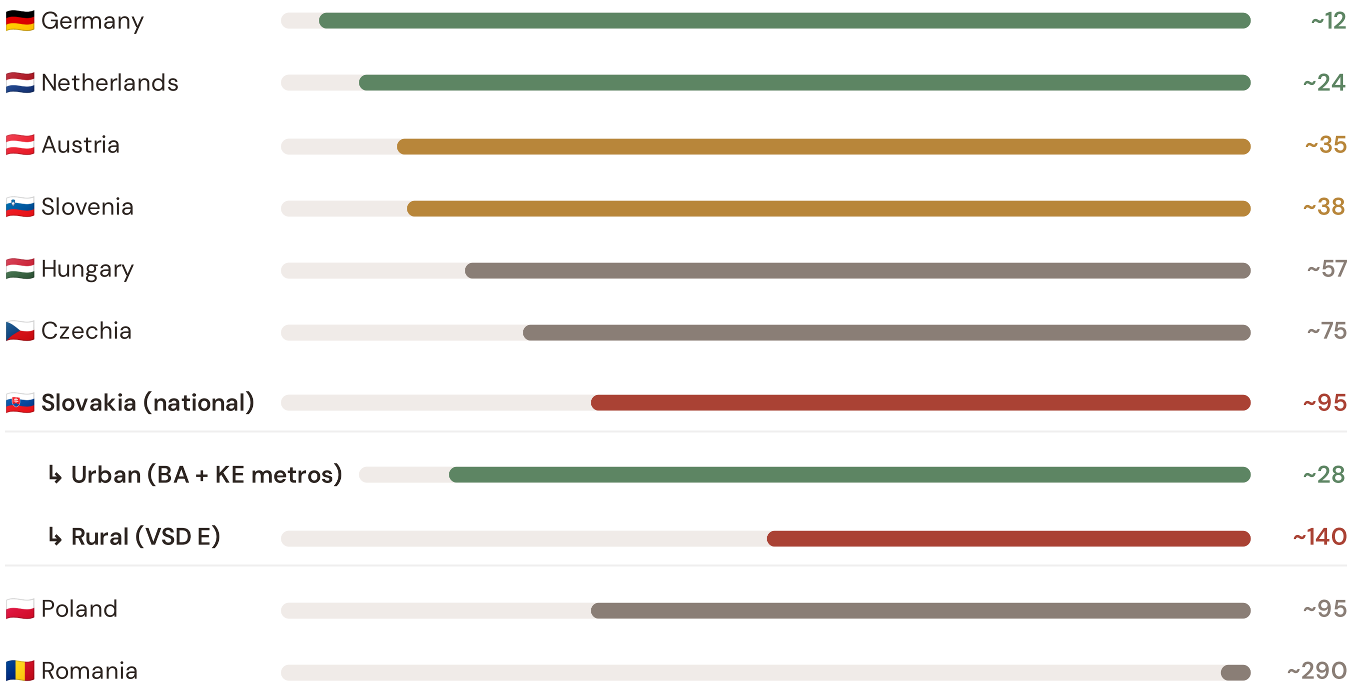
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Mochovce-corridor analysis hinges on Slovakia's continuity performance relative to European peers, especially the Visegrád-4 + CEE-South cohort siblings. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs DACH peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 01 · July 2026

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



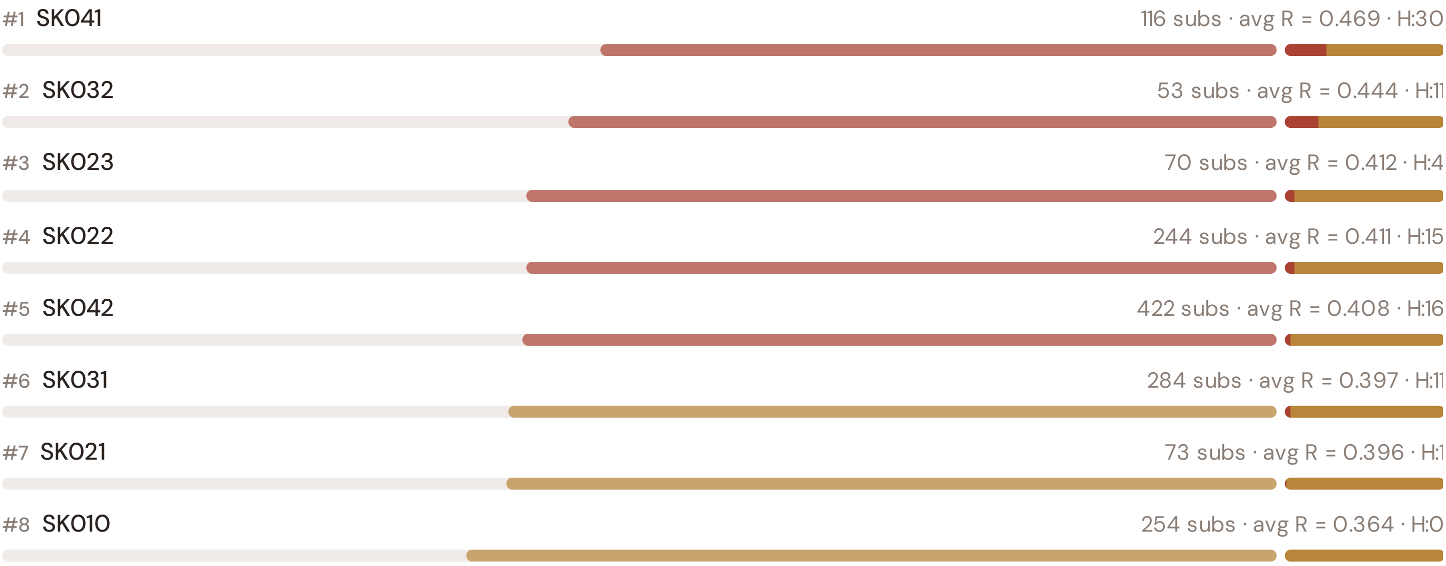
Source: CEER 7th Benchmarking Report (2022); ÚRSO 2023 quality reports; per-DSO filings (ZSD/SSD/VSD 2023). Slovakia sub-categories from ÚRSO territory classification (urban metros vs VSD eastern rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The Nitriansky kraj — Mochovce nuclear corridor corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0× the fleet average** of 0.475. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 8 Slovak NUTS-3 Kraje

Where does Nitriansky kraj sit within the national landscape? Kraje sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



The Nitriansky kraj — Mochovce nuclear corridor ranks **#0 of 8 nuts-3 regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are SK041, SK032, SK023, while the three most resilient are SK010, SK021, SK031. 5 of 8 score above the fleet average of 0.405. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

> **From grid intelligence to BESS valuation platform.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to ZSD/SSD/VSD and to SEPS — and to the community?"* Each edition explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data
SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation
DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net
GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage
Country prioritisation, DCI, registry, calibration

Layer 5 — Platform
Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 1,516 substations with complete feature vectors. Average CI_width of 0.073 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 1 Low, 1427 Medium, 88 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

The Central Highlands

Deep-dive into Banskobystrický kraj's grid risk profile — substation map, component analysis, nuts-3 regions breakdown. European Context Card: Žiar nad Hronom aluminium + Hron valley + Slovenský kras karst hydrogeology.

NEXT DATA REFRESH

Q3 2026 — 4 Quarterly Sources

DSO filings (ZSD/SSD/VSD), ŠÚ SR, NBS, ÚJD SR (nuclear) are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **ÚRSO — Úrad pre reguláciu sieťových odvetví** (12 variables → C1,C2,V1,V2,T1).

FLEET SPOTLIGHT

SK041: Highest Risk Concentration

SK041 has the highest concentration of High/Critical substations: **30 of 116 (26%)** are in the upper risk bands — avg R = 0.469. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Banskobystrický kraj**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Slovak utility, ÚRSO, SEPS, ZSD/SSD/VSD, ÚJD SR, Slovenské elektrárne, NBÚ, SK-CERT, or an academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Slovakia Edition 01 (Inaugural)

Published 9 July 2026 · SSI Index v4.0.2 · 1,516 substations · ~1,636 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5
